

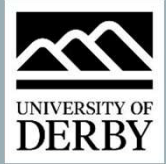
5CC506 Data Driven Systems

Introduction to Databases

Konstantinos Katsifis

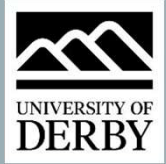


Academic Year 2024-2025



What is a database?

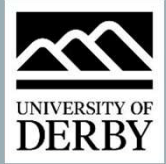
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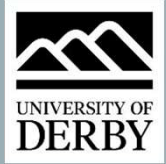
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A definition: a database is a persistent, logically coherent collection of inherently meaningful data, which capture the relations of real aspects in our world.



Why are data important?

People and businesses rely on data...

ACTIVITY: Think about an example relevant to your programme of study



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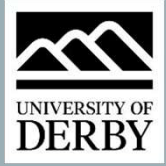
ACTIVITY: Think about an example relevant to your programme of study

- Enterprise Information (Sales, Accounting, Human Resources, ...)
- Banking and Finance (Banking, Transactions, Finance, ...)
- Universities
- Airlines
- Telecommunications



Purpose of DB Systems

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This had a number of major disadvantages:

- Data redundancy and inconsistency
- Difficulty in accessing data
- Data isolation
- Integrity problems
- Atomicity problems
- Concurrent-access anomalies
- Security problems

Data Abstraction

Three level of abstraction:

- View Level

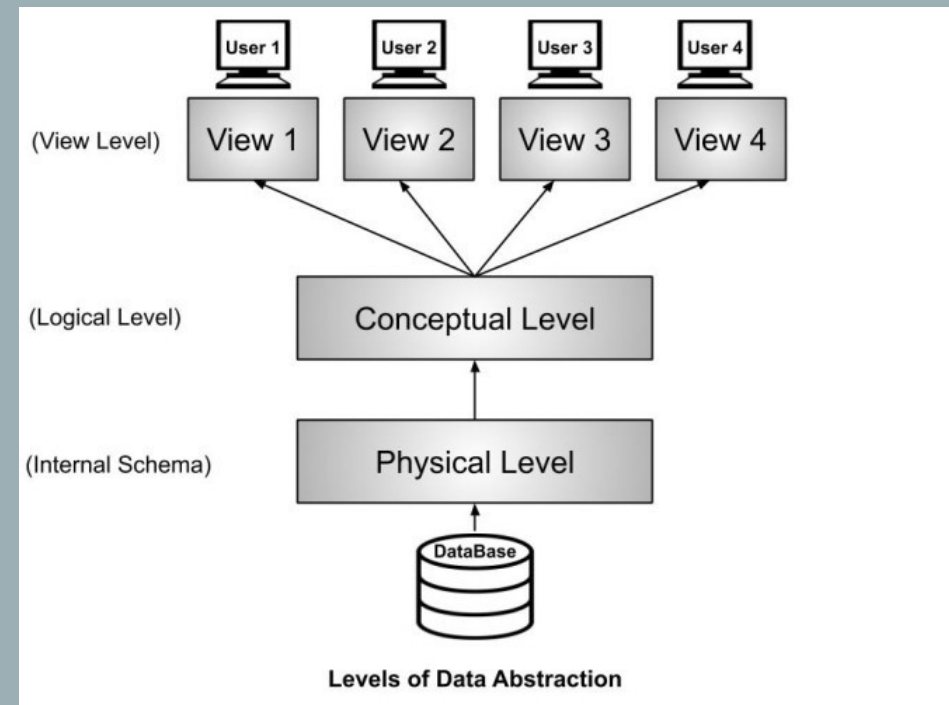
Displays part of the database

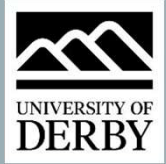
- Logical Level

What data are stored and how

- Physical Level

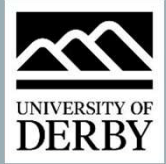
How the data are actually stored





Data Models

A **Data Model** is a collection of conceptual tools for describing data, data relationships, data semantics and consistency constraints.



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There are a number of different data models:

- *Relational Model* – uses a collection of tables to represent data and relationships among data
- *Entity-Relationship Model* – uses a collection of basic objects, entities, and relationships among these objects
- *Object-Based Data Model* – extends the E/R model with an approach typical of OOP, i.e. encapsulation, methods, etc.
- ...



Database Languages

- Data Manipulation Language (DML)

Enables users to access or manipulate data as organised by the Data Model, e.g. retrieval or modification of information, insertion of new information, etc.



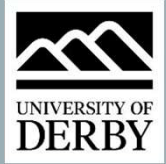
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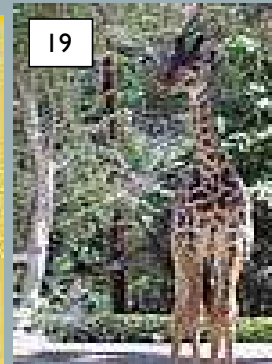
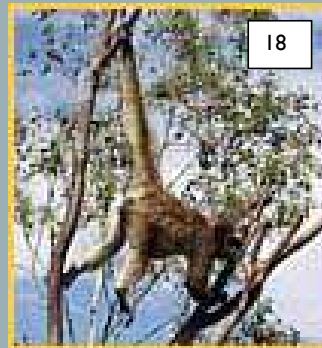
It is good to clarify that the above languages do NOT constitute separate languages but are part of the same language, e.g. SQL



Playing with data...

ACTIVITY: Consider the animal dataset that was part of the pre-session task for this week. You will now be split into break-out groups. In your group, identify all the possible grouping that you can find. Be original: if you find interesting groupings, feel free to note them down in the provided record sheet and explain the reason for each grouping. An example is provided

Duration of activity: **10 minutes**



1	Butterfly
2	Tiger
3	Tarantula spider
4	Sea Anemone
5	Scorpion
6	Blue Whale
7	Starfish
8	Mallard Duck
9	Crab
10	Gazelle
11	Lion
12	Gnu
13	Camel
14	Lizard
15	Fish
16	Parrot
17	Meerkat
18	Spider Monkey
19	Giraffe
20	Octopus



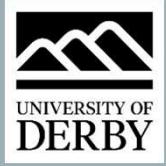
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Part 2: Each group challenges the other groups to identify the reason for one of their groupings by giving them the animal numbers that are part of the grouping



References

Silberschatz, A., Korth, H. F., and Sudarshan, S. (2010), *Database System Concepts*, 6th Edition. McGraw-Hill.

- Chapter 1 – Introduction (up to and including Section 1.4)